

Assignment 2

Research methods and tools

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Smart traffic light control

Part 1

For the deeply analysis of the theme we used 5W methodology. This approach made it able to systematize the key aspects of the problem.

Who? Drivers, pedestrians, cyclist, emergency services, urban planers and BigTech companies.

What? Was analyzed factors like, travel time, safety, fuel consumption and air pollution levels.

When? The topic is essentially relevant today, with the growth of active development of the metropolis, their population is also growing accordingly, which is why the presence of smart traffic management systems is so important.

Where? Huge cities and megapolises with the high traffic density.

Why? The importance of the topic is due to the need to solve traffic congestion problems, improve road safety and reduce the negative impact on the environment.

How the implementation of artificial intelligence algorithms in traffic light control systems in cities and metropolitan areas can reduce traffic jam times during rush hour?

Part 2

Research aim: The aim of the study is to investigate and evaluate the effectiveness of existing artificial intelligence-based traffic light control systems in reducing the average waiting time in traffic jams during peak periods through an experiment using existing models and research

Research objectives:

- First, conduct a literature review, i.e., study existing works on smart traffic management systems, in order to identify key methodologies, issues, and metrics.
- Secondly, a search for existing models and works to analyze their traffic flow dynamics during rush hour, as well as an analysis of their traffic light control systems and algorithms and a comparison of their performance

- Thirdly, an assessment of the effectiveness of various artificial intelligence systems based on indicators such as average waiting time, queue length, and throughput to confirm or refute the hypothesis

Research gaps:

- Firstly, the lack of quantitative evaluation of specific algorithms during rush hour. Many studies confirm the effectiveness of smart traffic lights, but they often lack a detailed analysis of performance under peak load conditions. This study aims to fill this gap by comparing various algorithms.
- Secondly, limited adaptability: some systems operate on the basis of predefined scenarios and not fully dynamic real-time learning. Systems with real-time learning assume higher efficiency.
- Thirdly, the problem of transferring the results to the real world, since there may be gaps in implementation in standard modeling tools and the research data is aimed at creating the most realistic scenario using proven

Part 3

Urban mobility optimization: Quantifying the impact of AI-based traffic light control on vehicle waiting times.

Part 4

Introduction

Definition of the terms:

Adaptive traffic light control(ATLC): A control system that adjusts traffic light phase durations in real time based on current situation data.

Reinforcement learning(RL): A field of machine learning in which a system learns to make decisions in an environment to maximize rewards.

Description of the problem:

Urbanization and the growth of transport in megacities have led to traffic jams becoming one of the main problems of modern cities. The current system of managing a fixed cycle is ineffective in conditions of dynamic flow - this leads to problems in the economic part and problems of environmental consequences such as increased CO2 emissions.

To address these issues, smart traffic systems have been developed, the core element of which is real-time traffic signal reading and control. ATLCS systems use real-time data from cameras to analyze the situation and make decisions. The most suitable approach is to use artificial intelligence with reinforcement learning, which helps the system learn from traffic data and develop an optimal strategy for solving the problem.

Research shows the high potential of such systems. Tests of adaptive systems have shown a reduction in waiting times at intersections by more than 40%. Simulations have shown that artificial intelligence-based systems can improve traffic flow efficiency and reduce emissions and fuel consumption.

Despite these results, there remains a need for a more detailed study of the effectiveness of specific artificial intelligence algorithms in more complex situations. To achieve this goal, the standard SUMO platform will be used, which will allow for a controlled experiment. This study aims to address this issue by conducting a simulation experiment and analyzing a traffic light control system based on reinforcement artificial intelligence for its ability to reduce average waiting times in traffic jams.

References

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